

Outset · AI in Healthcare · Day 3 · Group 6

Does estrogen keep the mind sharp?

A twenty-year lesson in confounding, and why a survey can hint but only an experiment can prove

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BACKGROUND

A twenty-year detective story

THE EXCITING CLUE (1996-2002)

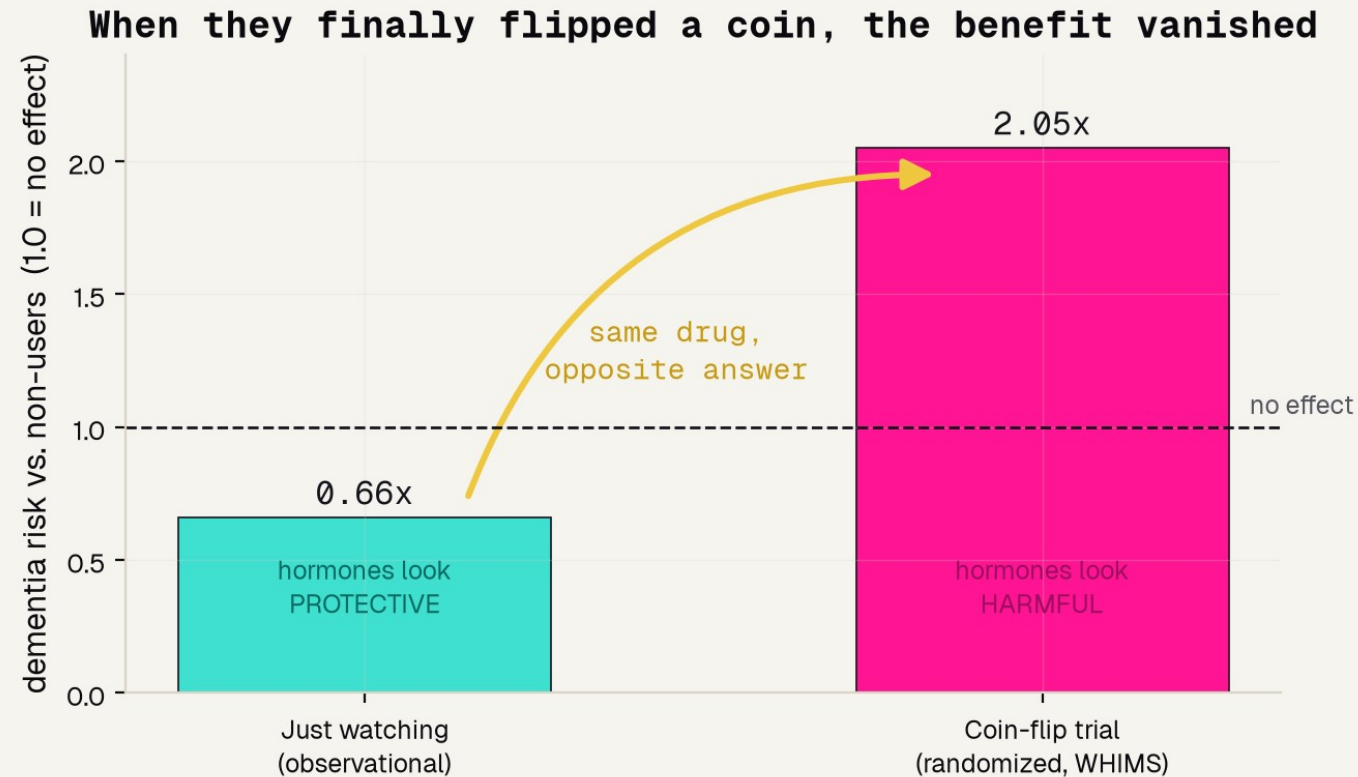
Studies that simply WATCHED women found that estrogen users developed Alzheimer's less often, and later, than women who never took it [1, 2]. For years this looked like solid proof that the pills protected the aging brain.

THE PLOT TWIST (2003)

Then the WHIMS trial let a COIN FLIP -- not the woman herself -- decide who got hormones. The memory benefit vanished, and dementia risk actually ROSE in women aged 65 and older [3, 4].

Same drug, opposite answer. The rest of this talk is one question: HOW?

When they flipped a coin, the benefit vanished

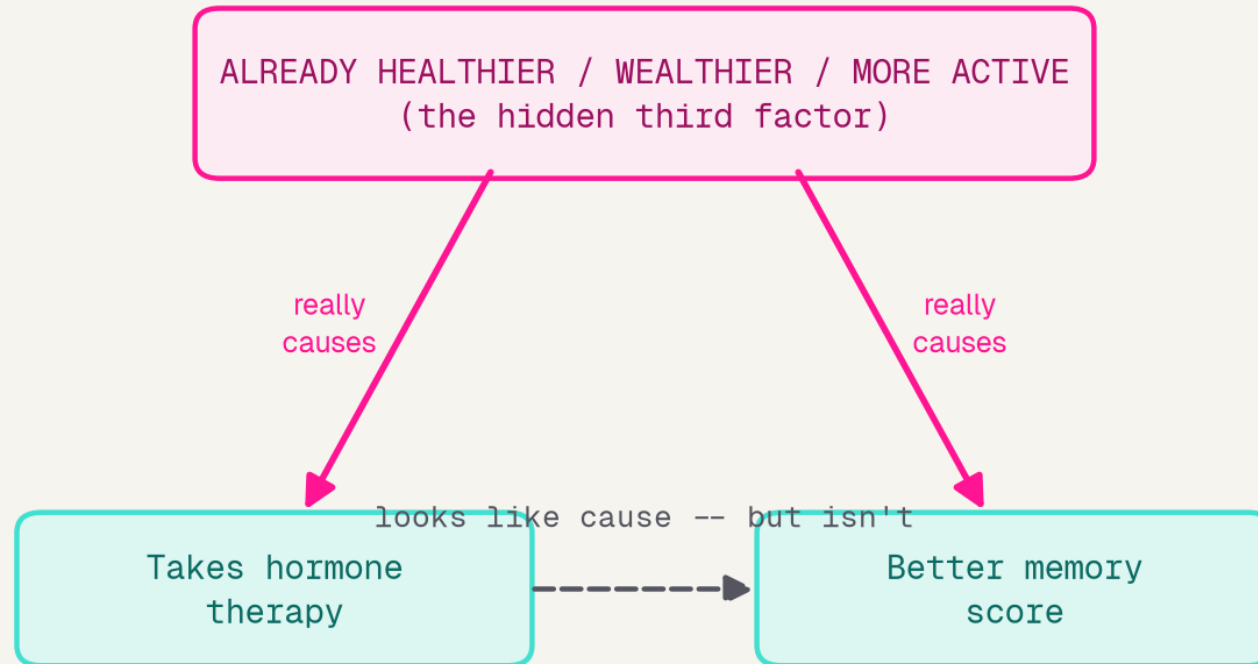


adapted from Tang 1996 / Zandi 2002 (observational) vs. Shumaker 2003 WHIMS (randomized)

Just watching (turquoise) said hormones PROTECT; the coin-flip trial (pink) said they HARM -- the study design, not the drug, decided the answer.

Healthy-user bias: a hidden third factor

Healthy-user bias: a hidden factor fakes the link



concept -- adapted from Wharton 2009 / Vandenbroucke 2009

Who chose the pill?

Users were richer, more active, and saw doctors more often to begin with [5].

The hidden factor

That same head start drives BOTH the drug use AND the good memory.

So the bottom link

only LOOKS like cause. It is a mirage -- confounding, a.k.a. healthy-user bias [6].

Ice cream and shark attacks

On hot days, ice-cream sales and shark attacks BOTH rise together. Does ice cream summon sharks?

THE ANALOGY

No -- the hot SUMMER WEATHER drives both at once: heat sells more ice cream AND sends more people into the ocean to swim. Neither one causes the other.

OUR VERSION

Estrogen use and good memory also rise together. But being HEALTHIER and better-off to start with can drive both -- so the pill itself may summon nothing.

When two things move together, hunt for the hidden third factor before you believe one caused the other.

METHODS

Our data: NHANES women 60 and older

739 WOMEN

all aged 60 or older, drawn from NHANES 2013-14 -- a real US government health survey, not data we made up.

USED ESTROGEN?

whether she ever took female hormones -- a simple yes or no, self-reported by each woman.

HER BACKGROUND

her age, education, and family income -- the confounders we have to account for.

MEMORY SCORE

a digit-symbol attention test; higher = sharper. Our model tries to flag the weaker-scoring half.

The one question we can actually ask

We cannot randomly ASSIGN estrogen in a survey -- so we can never prove cause here. But we can still ask a narrower, honest question:

Does the estrogen effect SURVIVE once we account for age, education, and income?

CRUDE

Estrogen's effect measured all by itself -- ignoring everything else about the woman.

ADJUSTED

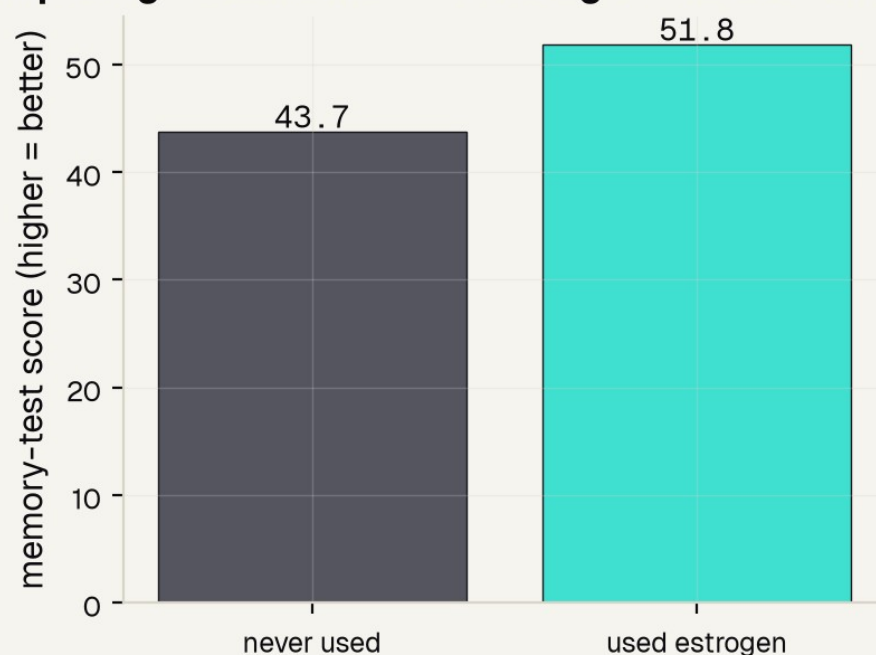
Estrogen's effect AFTER we account for her age, education, and income.

If the effect SHRINKS from crude to adjusted, the exciting headline was mostly confounding -- not real cause.

RESULTS

The tempting headline

The tempting headline: 'estrogen users think better'

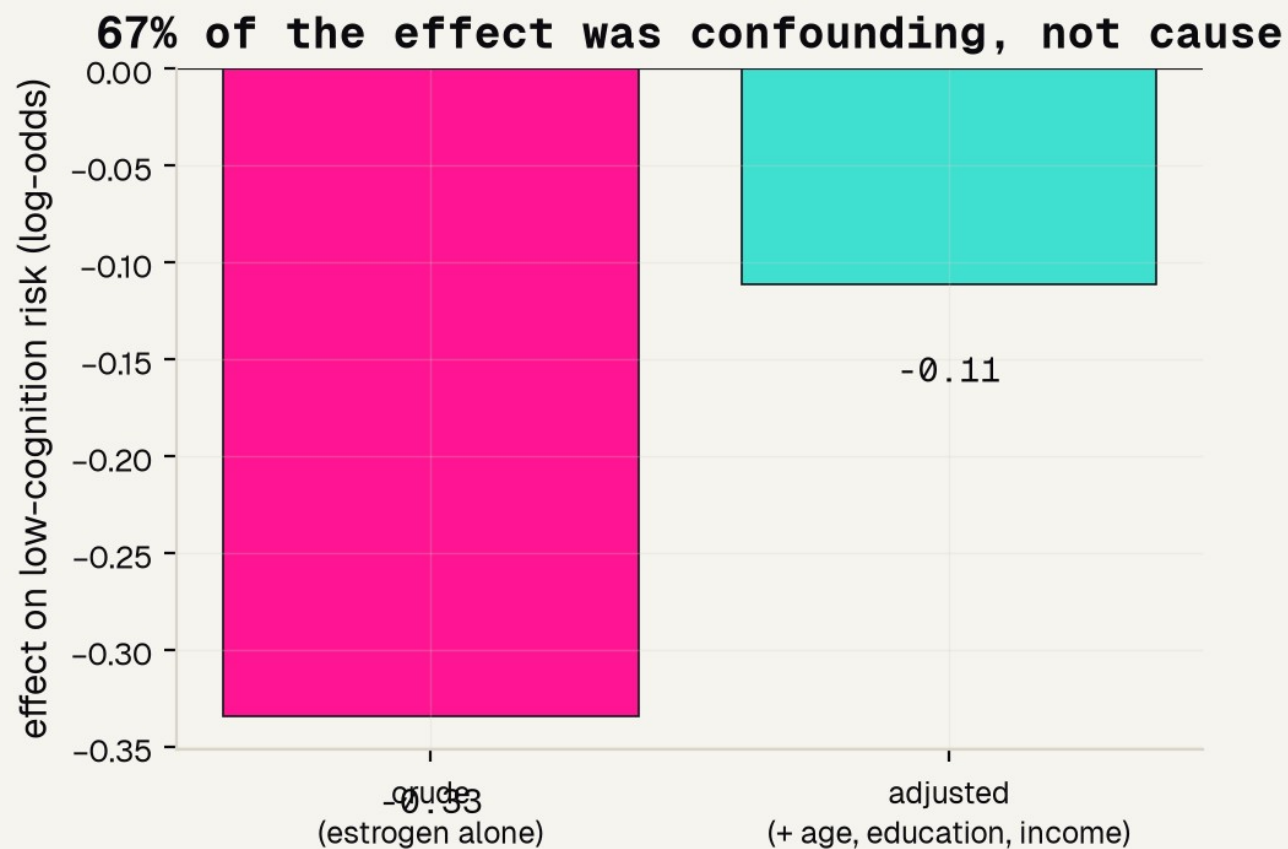


+8.1

points -- the raw gap, users over non-users

51.8 versus 43.7. Taken at face value, the exciting headline writes itself: 'estrogen users think better.' But we never checked whether the two groups were the same kind of people to begin with.

Two-thirds of the effect was confounding



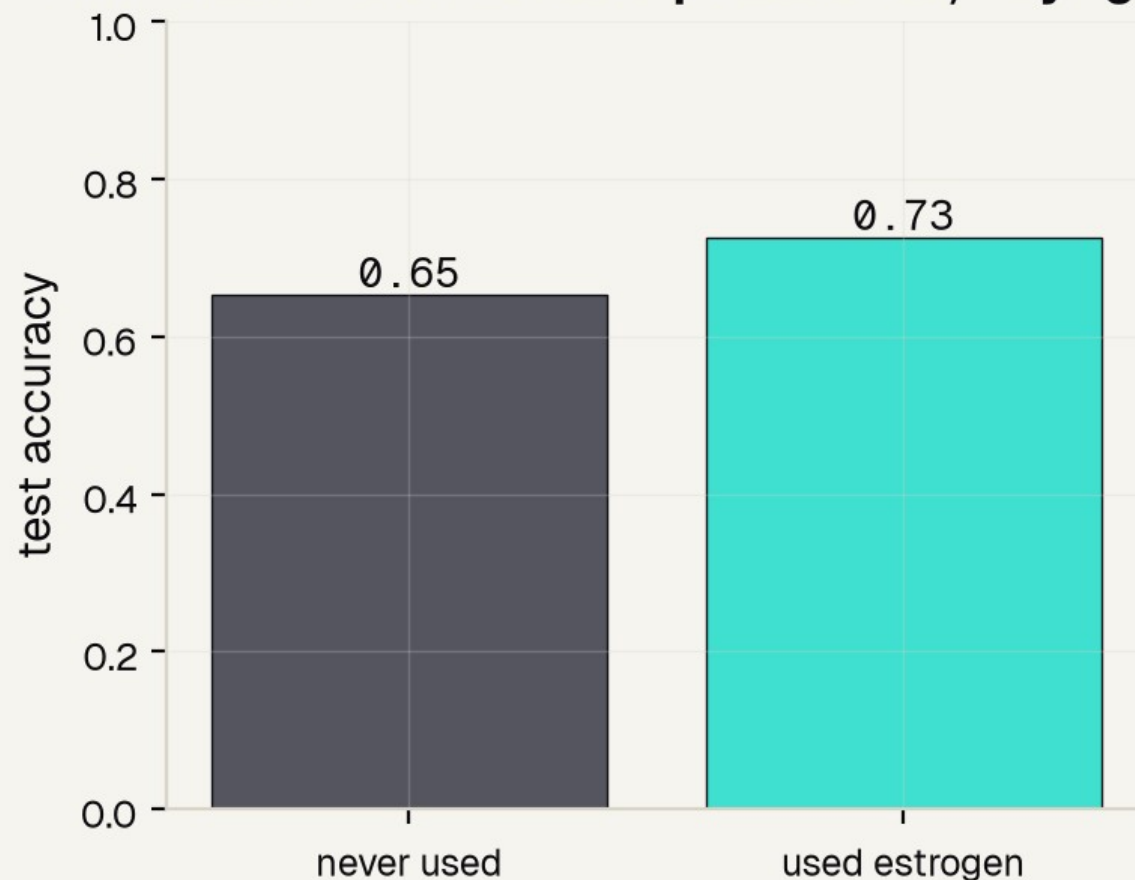
67%

of the effect was confounding

The crude effect (-0.33) shrank to just -0.11 once we adjusted for age, education, and income. That missing two-thirds was never the estrogen -- it was that users were healthier and better-off to start with.

Does the model work for everyone?

How well the model predicts, by group



0.73

test accuracy for estrogen-USERS

0.65

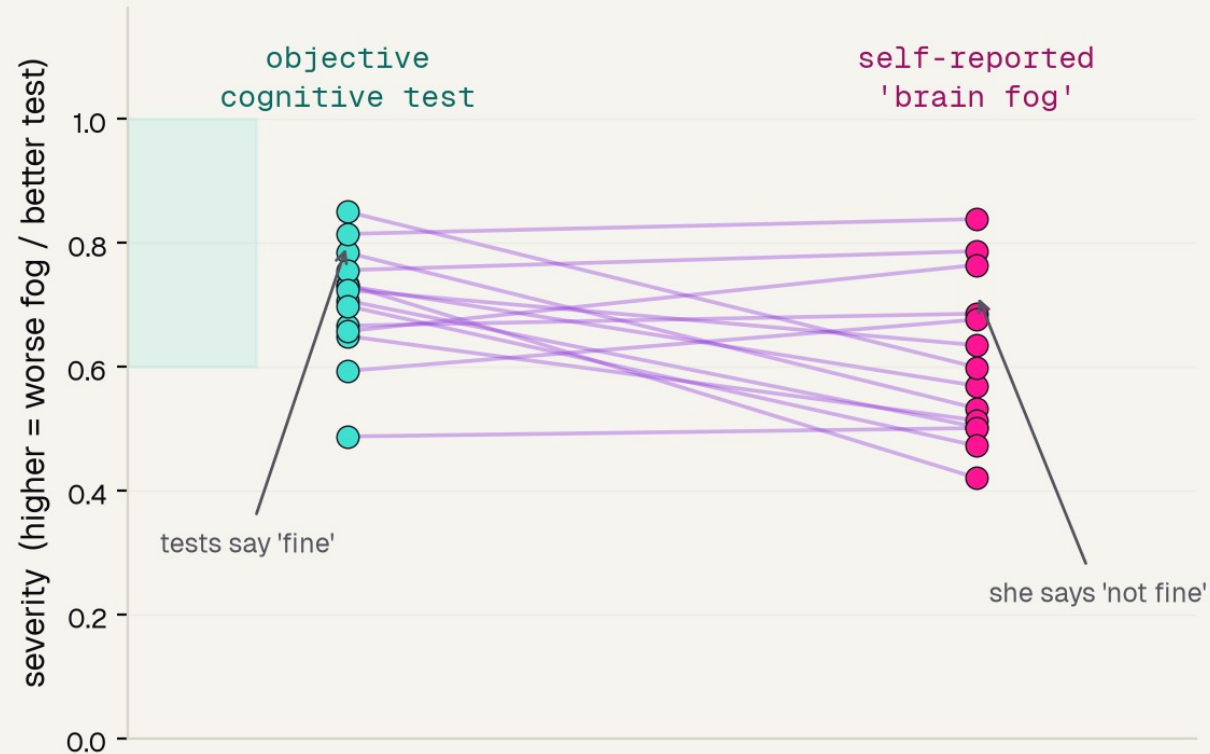
test accuracy for NON-users

A tool that is reliably right for one group and shakier for another can quietly do harm if a clinic trusts it equally. Naming the gap out loud is the honest move.

WHAT THE NUMBERS CAN'T CAPTURE

The tests say “fine”; she says “not fine”

The tests can say 'normal' while the patient feels foggy



adapted from Maki 2024, Harvard Health (menopause & brain fog)

The mismatch

Many women report real, daily brain fog even when the objective test looks perfectly normal [7].

Why it matters

Women's symptoms are too often waved off as stress or emotion instead of investigated [8].

Hold both ideas

The confounding is real AND the patient's account is real evidence -- not noise to dismiss.

A survey can hint; only an experiment proves

WHAT IT CAN DO

Show that a tempting headline is mostly a mirage, and that adjusting for background shrinks the apparent estrogen effect by about 67%.

WHAT IT CANNOT DO

Prove cause. We can only correct for the confounders we actually measured and thought of -- something unrecorded could still be doing the work.

WHAT SETTLES IT

A randomized trial: a coin flip, not the patient, assigns the drug, so the two groups start out identical. That is why WHIMS could overturn 20 years of headlines.

References

[1] Tang et al. 1996, Lancet -- observational; estrogen users got Alzheimer's less/later.

[2] Zandi et al. 2002, JAMA (Cache County) -- observational; long-term users lower risk.

[3] Shumaker et al. 2003, JAMA -- WHIMS RCT; hormones DOUBLED dementia risk in 65+.

[4] Rapp et al. 2003, JAMA -- WHIMS RCT; no cognitive benefit, scores trended worse.

[5] Wharton et al. 2009, Maturitas -- healthy-user bias; users start out healthier.

[6] Vandenbroucke 2009, Lancet -- why observational studies and RCTs disagreed.

[7] Maki 2024, Harvard Health -- menopause 'brain fog' vs. normal test scores.

[8] Chen & Shafir 2025, Harvard Health -- the dismissal of women's symptoms.

The one habit to carry out the door

Suspect the chooser, not the treatment.

When a group that CHOSE a treatment looks healthier, the head start usually belongs to the person, not the pill -- and only a coin-flip experiment can promote a correlation to a cause.

And take the patient's account seriously -- even when the test says 'fine.'

Thank You

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